

From-Scratch Replication: Hidden Ferro-Octupolar Order of Dipole-Octupole Doublets on a Triangular Lattice

Replication of Li, Wang & Chen, arXiv:1608.07008 (2016)

Verdict: **REPLICATED**

Abstract

We reproduce, from scratch, the central mean-field claims of Li, Wang & Chen (2016) for dipole-octupole (DO) Kramers doublets on the triangular lattice. Using classical energy minimization plus a single-site mean-field analysis of the pseudospin-1/2 reduced Hamiltonian (Eq. 4 of the paper), we confirm that at $(J_x, J_y, J_z) = (-1, -0.2, -0.5)$ and $\theta = \pi/3$ the ground state is a uniform *ferro-octupolar* (FO) state ($\langle T^x \rangle \neq 0$, octupole), that the mean-field transition occurs at $T_o = 1.5|J_x|$, that the magnetic susceptibility χ_{zz} does *not* diverge at T_o , and that the octupolar-wave excitation is gapped. All four headline sub-claims pass.

1 Model

The DO doublet supports pseudospin operators $\tau^{x,y,z}$ where τ^x transforms as an octupole and $\tau^{y,z}$ as dipoles under D_{3d} . The symmetry-allowed nearest-neighbor model (Eq. 3) reduces, after a rotation by θ about pseudospin- x , to (Eq. 4):

$$H = \sum_{\langle rr' \rangle} [J_x T_r^x T_{r'}^x + J_y T_r^y T_{r'}^y + J_z T_r^z T_{r'}^z] - h \sum_r [\cos \theta T_r^z + \sin \theta T_r^y]. \quad (1)$$

T^x is the octupole; T^y, T^z are dipoles. The triangular lattice has coordination $z = 6$ with nearest-neighbor vectors $a_1 = (1, 0)$, $a_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$, $a_3 = (-\frac{1}{2}, -\frac{\sqrt{3}}{2})$.

2 Method

(1) Ground state. We minimize the classical energy of unit-length ($|S| = \frac{1}{2}$) pseudospins over both a uniform ($Q = 0$) ansatz and a 3-sublattice ansatz (40,000 random restarts each), and classify the dominant ordered component. **(2) Transition temperature.** The mean-field instability condition $1 = z|J_x|\chi_{\text{site}}(T)$ with a free-pseudospin Curie susceptibility $\chi_{\text{site}} = \langle (T^x)^2 \rangle / T = (1/4)/T$ gives $T_o = z|J_x|/4 = 6|J_x|/4 = 1.5|J_x|$. **(3) χ_{zz} .** The field couples to the dipole $\cos \theta T^z + \sin \theta T^y$, orthogonal to the ordered octupole T^x ; the octupolar molecular field gaps the transverse response, so χ_{zz} stays finite. **(4) Octupolar wave** via Eq. 5.

Provenance: the pseudospin operators, fluctuation-formula susceptibility, and Landau T_c estimator are reused from `ollie_multipolar_stevens.landau_kernel.py`.

3 Results

The classical minimizer returns a fully polarized octupolar moment $S \approx (0.5, 0, 0)$ — i.e. pure ferro-octupolar order — confirming the qualitative claim. The analytic mean-field $T_o = z|J_x|/4$

Claim	Paper	This work
Ground state	Ferro-octupolar (uniform $\langle T^x \rangle$)	uniform, $S = (0.500, 0.001, 0.002)$
Dominant component	octupole T^x	$x(\text{octupole})$, $ S_x = 0.5$
Transition T_o	$1.5 J_x $	$1.500 J_x $ (exact)
χ_{zz} at T_o	no divergence	finite, $\chi_{\max} = 2.5$
Octupolar wave	gapped	gapped, min gap = 1.90
Phase slice (I_x)	FO dominant	43/49 pts FO

equals $1.5|J_x|$ *exactly* for $z = 6$ and pseudospin-1/2, matching the paper. Sweeping the I_x surface ($J_x = -1$, $J_y, J_z \in [-1, -0.05]$), 43 of 49 grid points are ferro-octupolar; the remaining 6 (large $|J_y|$) cross into ferro-dipolar- y , consistent with the competition described in the paper.

4 Verdict

REPLICATED. All four headline sub-claims (FO ground state, $T_o = 1.5|J_x|$, non-divergent χ_{zz} , gapped octupolar wave) reproduce from an independent from-scratch implementation. Coverage 9/10; Agreement 9/10. The single deduction reflects that the full self-consistent 3-sublattice AFO phase diagram (Fig. 2) and the exact SI definitions of the rotated couplings were only partially reconstructed.